
JAL2301

DATA SCIENCE

UNIT – 1

INTRODUCTION TO DATA SCIENCE

SYLLABUS

UNIT – 1 INTRODUCTION TO DATA SCIENCE

- Data Science: Benefits and uses – Facets of data - Data Science Process - The big data ecosystem and data science - Overview of the data science process – Defining research goals – Retrieving data – Data preparation - Exploratory Data analysis – Build the models – Presenting findings and building applications - Basic Statistical descriptions of data.

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- <https://www.youtube.com/watch?v=nh4RgxaiKgI>
 - **Data Scientist Roles and Responsibilities | Data Scientist Career | Data Science Training | Edureka**



DATA SCIENCE



DATA SCIENCE

- ❑ **Big data** → any collection of data sets so large or complex that it becomes difficult to process them using traditional data management techniques (like the RDBMS [relational database management systems]).
- ❑ Characteristics of Big Data: **Three Vs**:
 - ❑ Volume - How much data is there?
 - ❑ Variety - How diverse are different types of data?
 - ❑ Velocity - At what speed is new data generated?
 - ❑ Veracity - How accurate is the data? → 4th V

DATA SCIENCE

- ❑ **Data science** → involves using methods to analyze massive amounts of data and extract the knowledge it contains.
- ❑ **Data science** → math + statistics + specialized programming + advanced analytics + AI + ML + subject-specific expertise
- ❑ **Data science** → combines math and statistics, specialized programming, advanced analytics, artificial intelligence (AI) and machine learning with specific subject matter expertise to uncover actionable insights hidden in an organization's data. These insights can be used to guide decision making and strategic planning.

DATA SCIENCE



DATA SCIENCE

- ❑ **Definition:** Data science is the study of data to extract meaningful insights for business. It is a multidisciplinary approach that combines principles and practices from the fields of mathematics, statistics, artificial intelligence, and computer engineering to analyze large amounts of data.
- ❑ **Data scientist** → A data scientist uses data to understand and explain the phenomena around them, and help organizations make better decisions → rely on popular programming languages and technologies to conduct exploratory data analysis and statistical regression → use Java, R Studio and Python and technologies like Hadoop, Pig, Spark.

DATA SCIENCE

Use cases: Few cases where DS was used

- ❑ An international bank - faster loan services with a mobile app
- ❑ A digital media technology company - TV audience analytics – insights into viewer behaviour
- ❑ An urban police department – locations to resources in order to prevent crime
- ❑ Shanghai Changjiang Science and Technology Development - AI-based medical assessment platform - analyze medical records - categorize patients based on their risk of experiencing a stroke - predict the success rate of different treatment plans

BENEFITS AND USES OF DATA SCIENCE

BENEFITS OF DATA SCIENCE

- ❑ **Commercial companies** - in almost every industry - use data science and big data
 - ❑ To gain insights into their customers, processes, staff, completion, and products
 - ❑ To offer customers a better user experience
 - ❑ To cross-sell, up-sell, and personalize their offerings
- Examples:
 - Google AdSense -  collects data from internet users so relevant commercial messages can be matched to the person browsing the internet.
 - MaxPoint → real-time personalized advertising

BENEFITS OF DATA SCIENCE

- ❑ **Human Resource Professionals** → use people analytics and text mining to screen candidates, monitor the mood of employees, and study informal networks among coworkers.
- ❑ **Financial Institutions** → use data science to predict stock markets, determine the risk of lending money, and learn how to attract new clients for their service.
- ❑ **Governmental Organizations** → rely on internal data scientists to discover valuable information and also share their data with the public → detecting fraud and other criminal activity or optimizing project funding. Eg. to monitor millions of individuals - collected 5 billion data records from widespread applications such as Google Maps, Angry Birds, email, and text messages, among many other data sources

BENEFITS OF DATA SCIENCE

- ❑ **Non-governmental Organizations (NGO)** → use data science to raise money and defend their causes → eg. WWF
 - ❑ Supported by data scientists – eg. group → DataKind
- ❑ **Universities** → use data science in their research but also to enhance the study experience of their students and to study how traditional classes and online courses compete.

FACETS OF DATA

FACETS OF DATA

- Different categories of data – each of them require different tools and techniques.
 - Structured data
 - Unstructured data
 - Natural language data
 - Machine-generated data
 - Graph-based data
 - Audio, video, and image data
 - Streaming data

FACETS OF DATA

Structured data

- Data that depends on a data model and resides in a fixed field within a record.
- Structured data stored in tables within databases or Excel files
- SQL, or Structured Query Language → the preferred way to manage and query data that resides in databases.

Unstructured data

- The content is context-specific or varying.
- Not easy to fit into a data model.
- Example: email



FACETS OF DATA



Natural language data

- A special type of unstructured data
- Challenging to process because it requires knowledge of specific data science techniques and linguistics
- Natural language processing → performs
 - summarization,
 - text completion,
 - sentiment analysis
- But deciphering every piece of text is challenging because natural language is ambiguous by nature (eg. bat, bank)
- entity recognition,
- topic recognition,



FACETS OF DATA

Machine-generated data

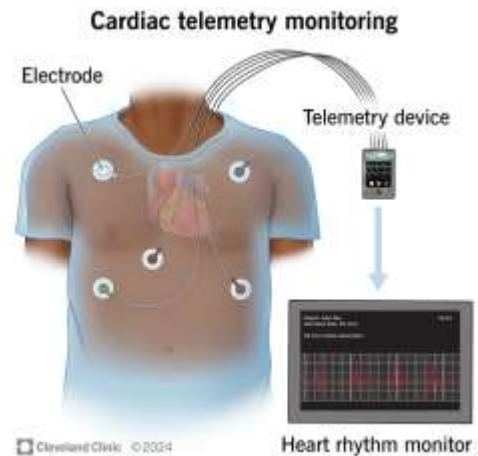
- Information that's automatically created by a computer, process, application, or other machine without human intervention.
- A major data resource
- High volume and speed → so analysis needs highly scalable tools.
- Internet of things (IoT) data
- Examples: web server logs, call detail records, network event logs, telemetry

FACETS OF DATA

Machine-generated data



Buoys



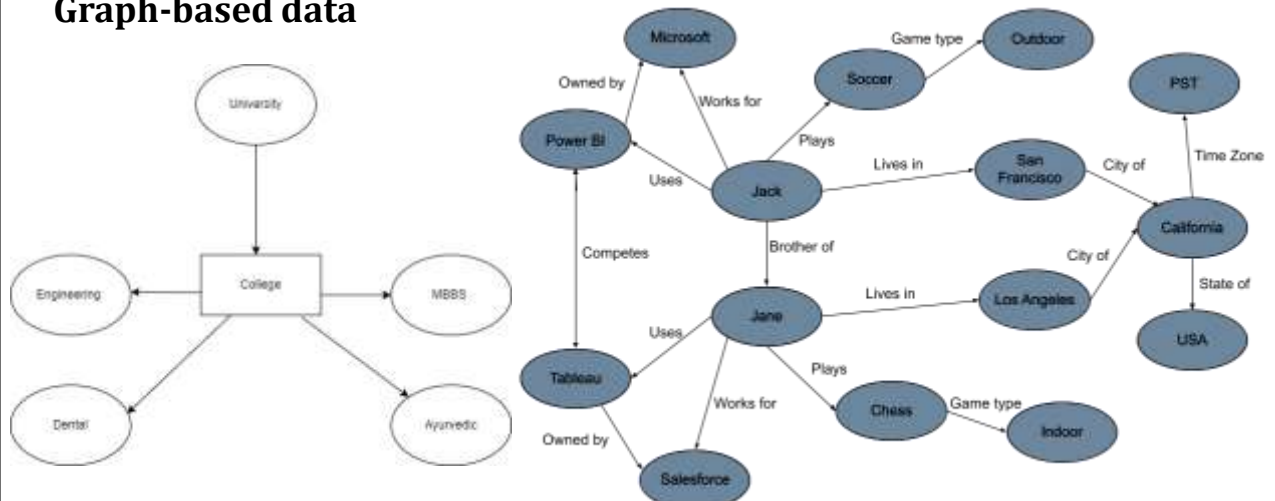
FACETS OF DATA

Graph-based data

- Graph → a mathematical structure to model pair-wise relationships between objects - use nodes, edges, and properties to represent and store graphical data.
- Graph-based data → a natural way to represent social networks → has a structure that allows us to calculate specific metrics (Example: the influence of a person and the shortest path between two people, facebook friends)
- Graph databases → used to store graph-based data -- queried with specialized query languages such as SPARQL.
- Graph data – challenging to handle, especially computer interpreting additive and image data

FACETS OF DATA

Graph-based data



FACETS OF DATA

Audio, video, and image data

- Highly challenging category of data
- Eg. object recognition, voice recognition
- Data captured by high speed cameras – consume more space

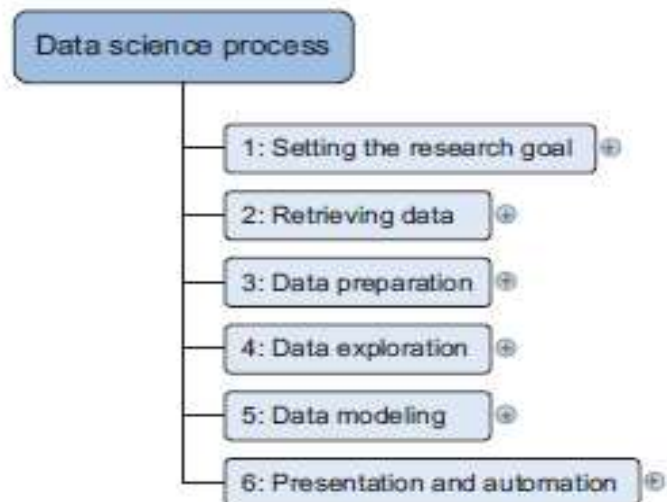
Streaming data

- Take any of the previous forms
- The data flows into the system when an event happens instead of being loaded into a data store in a batch
- Examples: “What’s trending” on Twitter, live sporting or music events, and the stock market, Online Gaming, IoT devices

DATA SCIENCE PROCESS

DATA SCIENCE PROCESS

- A six-step process



DATA SCIENCE PROCESS

Setting the research goal

- Prepare a project charter in the context of an organization
- What you're going to research
- How the company benefits from that
- What data and resources you need
- A timetable and
- Deliverables.

Retrieving data

- Collecting data as per the project charter
- Ensuring and checking the existence of, quality, and access to the data
- By third-party companies
- In any form

Data preparation

- enhance the quality of the
- data and prepare it for use in subsequent steps.
- data cleansing
- data integration
- data transformation

DATA SCIENCE PROCESS

- Data cleansing → removes false values from a data source and inconsistencies across data sources
- Data integration → enriches data sources by combining information from multiple data sources
- Data transformation → ensures that the data is in a suitable format for use in your models.

DATA SCIENCE PROCESS

Exploratory Data analysis (EDA)

- building a deeper understanding of the collected data
- how variables interact with each other;
- the distribution of the data
- whether there are outliers

Build the model

- use models, domain knowledge, and insights about the data found in the previous steps
- to answer the research question.
- select a technique from the fields of statistics, ML, operations research, and so on.
- an iterative process that
- involves selecting the variables for the model, executing the model, and model diagnostics.

Presenting findings and building applications

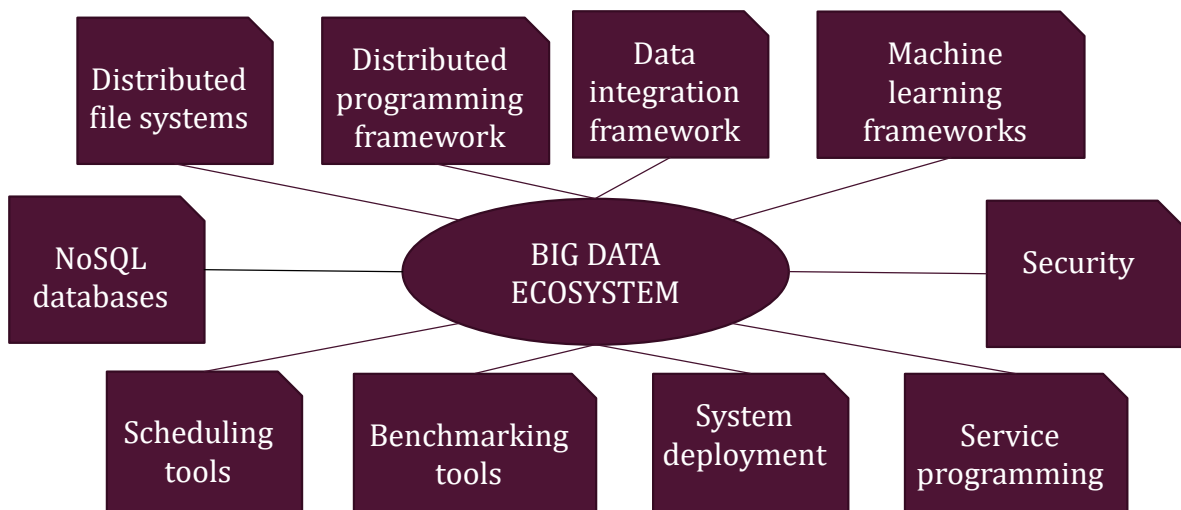
- present the results to your business
- Presentations, graphs / charts, research reports
- Must be an automated process

THE BIG DATA ECOSYSTEM AND DATA SCIENCE

BIG DATA ECOSYSTEM AND DATA SCIENCE

- ❑ Big data ecosystem → big data tools and frameworks
- ❑ New technologies appear rapidly.
- ❑ Data scientists use many different technologies,
- ❑ Technologies that have similar goals and functionalities can be grouped in big data ecosystem

BIG DATA ECOSYSTEM AND DATA SCIENCE



BIG DATA ECOSYSTEM AND DATA SCIENCE

Distributed File Systems (DFS)

- ❑ Similar to a normal file system
- ❑ **Advantages:**
 - Can store files larger than any one computer disk.
 - Automatic replication of files across multiple servers for redundancy or parallel operations
 - Scalable system
- ❑ Earlier – vertical scaling (moving contents to a new server) ; nowadays – horizontal scaling (adding another small server)
- ❑ Eg. HDFS (for Google file system), Red Hat Cluster File System, Ceph File System, and Tachyon File System,

BIG DATA ECOSYSTEM AND DATA SCIENCE

Distributed programming framework

- ❑ Needed to exploit DFS to get much better experience working with distributed data
- ❑ Programs / applications designed using normal programming languages not suitable to deal with complexities.
- ❑ Eg. Apache Hadoop, Apache Spark, Apache Flink, and Apache Storm

Data integration framework

- ❑ Adding and moving data possible through data integration frameworks such as Apache Sqoop and Apache Flume excel.
- ❑ Similar to ETL process (extract, transform, and load process) in a traditional data warehouse.

BIG DATA ECOSYSTEM AND DATA SCIENCE

Machine Learning Frameworks

- ❑ To extract the coveted insights from the collected and stored data – using computers
- ❑ Mathematical formulas, write them in an algorithm,
- ❑ More number of computers for analysis – with time complexity
- ❑ Specialized frameworks and libraries are required to deal with this amount of data.
- ❑ Python is Scikit-learn – a great machine-learning toolbox for ML
- ❑ Other Python libraries → PyBrain for neural networks, NLTK or Natural Language Toolkit, Pylearn2, TensorFlow

BIG DATA ECOSYSTEM AND DATA SCIENCE

NoSQL Databases

- ❑ Huge amounts of data- require software that's specialized in managing and querying this data
- ❑ Traditional relational databases → Oracle SQL, MySQL, Sybase→
- ❑ Issues → inadequate scalability with storage or processing power, handling streaming, graph, or unstructured forms of data,
- ❑ New type → NoSQL databases → 'Not only' SQL
- ❑ Categories of new databases:

BIG DATA ECOSYSTEM AND DATA SCIENCE

NoSQL Databases: Categories of new databases:

TYPE OF DATABASE	DESCRIPTION
Column databases	Data is stored in columns, which allows algorithms to perform much faster queries. Newer technologies use cell-wise storage. Table- like structures are still important.
Document stores	Document stores no longer use tables, but store every observation in a document. This allows for a much more flexible data scheme.
Streaming data	Data is collected, transformed, and aggregated not in batches but in real time. It's more a particular type of problem that drove creation of technologies such as Storm.

BIG DATA ECOSYSTEM AND DATA SCIENCE

TYPE OF DATABASE	DESCRIPTION
Key-value stores	Data isn't stored in a table; rather you assign a key for every value, Eg. - org.marketing.sales.2015: 20000. This scales well but places almost all the implementation on the developer.
SQL on Hadoop	Batch queries on Hadoop are in a SQL-like language that uses the map- reduce framework in the background.
New SQL	This class combines the scalability of NoSQL databases with the advantages of relational databases. They all have a SQL interface and a relational data model.
Graph databases	Not every problem is best stored in a table. Particular problems are more naturally translated into graph theory and stored in graph

BIG DATA ECOSYSTEM AND DATA SCIENCE

Scheduling tools

- ❑ To automate repetitive tasks and trigger jobs based on events such as adding a new file to a folder.
- ❑ Eg. tools CRON on Linux (specifically developed for big data).
- ❑ Eg. MapReduce task

Benchmarking tools

- ❑ To optimize big data installation by providing standardized profiling suites.
- ❑ Using an optimized infrastructure can make a big cost difference.
- ❑ For example, if you can gain 10% on a cluster of 100 servers, you save the cost of 10 servers.

BIG DATA ECOSYSTEM AND DATA SCIENCE

System deployment

- ❑ Setting up a big data infrastructure – not an easy task
- ❑ Help in automating the installation and configuration of big data components.

Service programming

- ❑ Service tools excel by exposing big data applications (eg. a prediction application on Hadoop) to other applications as a service.
 - Eg. REST service (representational state transfer → used to feed websites with data)

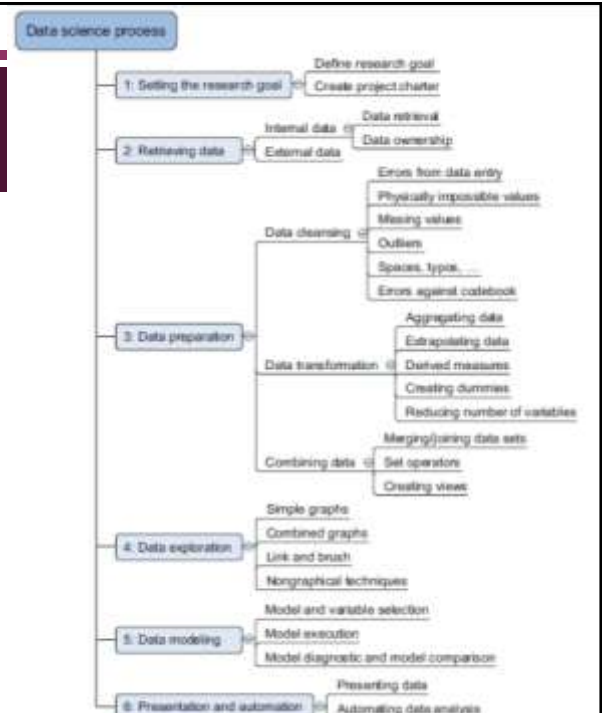
Security

- ❑ Big data security tools allow you to have central and fine-grained control over access to the data.
- ❑ By security experts.

OVERVIEW OF THE DATA SCIENCE PROCESS

OVERVIEW OF THE DATA SCIENCE PROCESS

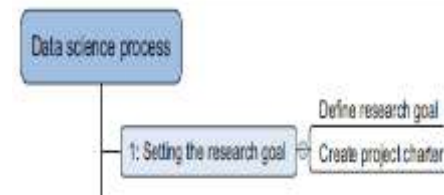
- A structured approach followed in data science – hence chances of success maximized in a data science project at the lowest cost
- Project worked as a team



OVERVIEW OF THE DATA SCIENCE PROCESS

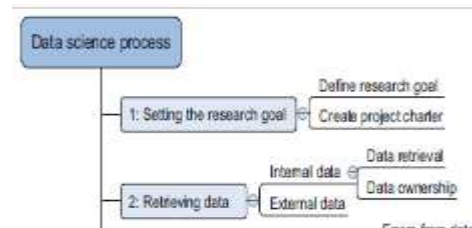
❑ Step-1: Setting the research goal

- Purpose: For making sure all the stakeholders understand the what, how, and why of the project.
- Project charter prepared



❑ Step-2: Data Retrieval

- Finding suitable data and getting access to the data from the data owner.
- Result → data in its raw form → needs to be transformed to usable form

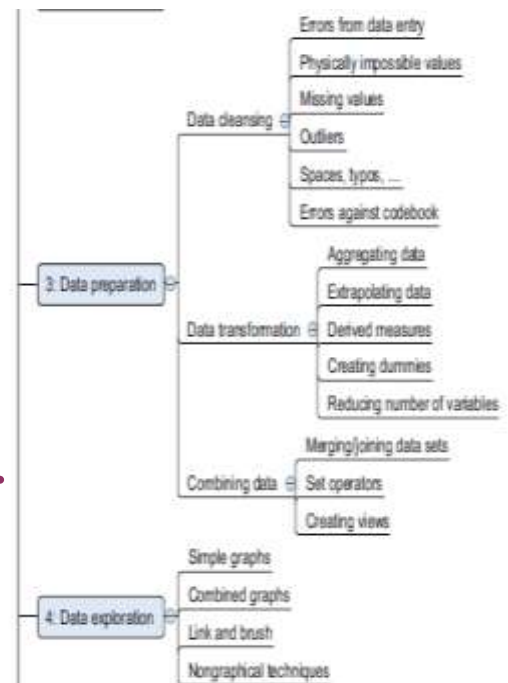


OVERVIEW OF THE DATA SCIENCE PROCESS

- Data in raw form is transformed to usable form.
- Detect and correct different kinds of errors in the data,
- Combine data from different data sources, and transform it.

❑ Step-3: Data Preparation

- Goal: to gain a deep understanding of the data.
- Look for patterns, correlations, and deviations based on visual and descriptive



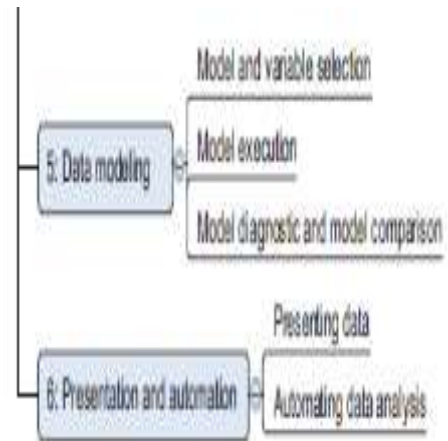
OVERVIEW OF THE DATA SCIENCE PROCESS

❑ **Step-5: Model Building**

- Gain the insights or make the predictions stated in the project charter.
- A combination of simple models may outperform one complicated model

❑ **Step-6: Presentation and Automation**

- Convince the business that your findings will indeed change the business process as expected.
- Strategic and tactical approach needed
- Automating the project will save time.



DATA SCIENCE PROCESS

- ❑ Defining research goals
- ❑ Retrieving data
- ❑ Data preparation
- ❑ Exploratory Data analysis
- ❑ Build the model
- ❑ Presenting findings and building applications

1. DEFINING RESEARCH GOALS

STEP-1: DEFINING RESEARCH GOALS AND CREATING A PROJECT CHARTER

- ❑ Goal: answering the questions – what, why and how?
 - What does the company expect you to do?
 - Why does management place such a value on your research?
 - Is it part of a bigger strategic picture or a “lone wolf” project originating from an opportunity someone detected?
- ❑ Then everybody knows what to do and can agree on the best course of action.
- ❑ Outcome:
 - a clear research goal,
 - a good understanding of the context,
 - well-defined deliverables, and
 - a plan of action with a timetable.
- ❑ This information is then best placed in a **project charter**.

STEP-1: DEFINING RESEARCH GOALS AND CREATING A PROJECT CHARTER

~~Spend time understanding the goals and context of your research~~

- ❑ Understanding the business goals and context is critical for project success.
- ❑ Continue asking questions and devising examples until you grasp the exact business expectations.
- ❑ If not, then after few months of research everyone immediately realizes that you misunderstood their question.

Create a project charter

- ❑ Clients like to know what they're

- ❑ So after understanding the business problem, get a formal agreement on the deliverables – in a project charter with the following:

1. A clear research goal
2. The project mission and context
3. How you're going to perform your analysis
4. What resources you expect to use
5. Proof that it's an achievable project, or proof of concepts
6. Deliverables and a measure of success
7. A timeline

2. RETRIEVING DATA

STEP-2: RETRIEVING DATA

- ❑ Sometimes it is needed to go into the field and design a data collection process.
- ❑ Many companies will have already collected and stored the data,
- ❑ If they don't have data, it can often be bought from third parties.
- ❑ Data can be stored in many forms, ranging from simple text files to tables in a database.
- ❑ The objective → acquiring all the data needed → may be a difficult process.

STEP-2: RETRIEVING DATA

Start with data stored within the company

- ❑ Data can be stored in official data repositories such as databases (data storage), data warehouses (for reading and analyzing that data), data marts (subset of the data warehouse – business-specific), and data lakes (raw data) maintained by a team of IT professionals.

Don't be afraid to shop around

- ❑ If data isn't available inside the organization, look outside the organization's walls
- ❑ Eg. Twitter, LinkedIn, and Facebook collect data

Do data quality checks now to prevent problems later

- ❑ Compare source document and the data for similarity – this will reduce errors hence data preparation step will be easier

3. DATA PREPARATION

➤ *CLEANSING DATA*

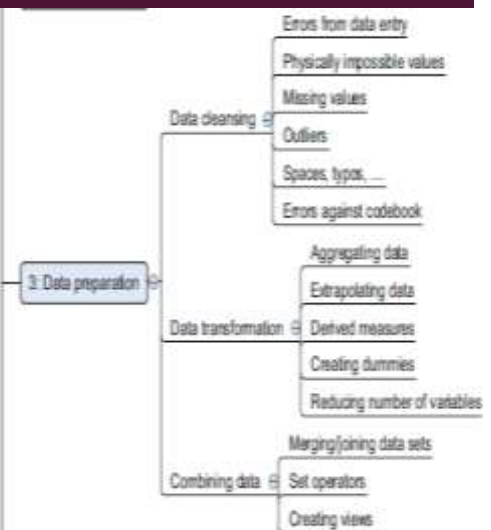
➤ *CORRECT ERRORS AS EARLY AS POSSIBLE*

➤ *COMBINING DATA FROM DIFFERENT DATA SOURCES*

➤ *TRANSFORMING DATA*

DATA PREPARATION

- ❑ The raw data received from the data retrieval phase is likely to be sanitized and prepared for use in the modeling and reporting phase – as per the specific format needed by the model.
- ❑ It is better to correct errors as early as possible.
- ❑ Common actions in data preparation:
 - Cleansing data
 - Integrating data
 - Transforming data



DATA PREPARATION CLEANSING DATA

- ❑ Data cleansing → a subprocess of the data science process that focuses on removing errors in your data so your data becomes a true and consistent representation of the processes it originates from.
- ❑ Two types of errors exist.
 - **Interpretation error** → Eg. person's age > 300
 - **Inconsistencies** → between data sources or against the company's standardized values → eg. gender is 'Male' in one table and 'M' in another, currency is in \$ and €.
- ❑ Few more general errors:

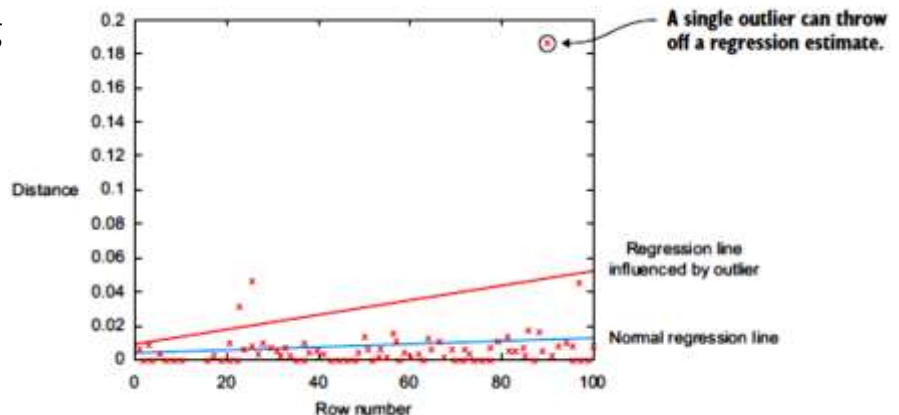
DATA PREPARATION CLEANSING DATA

Error description	Possible solution
<i>Errors pointing to false values within one data set</i>	
Mistakes during data entry	Manual overrules
Redundant white space	Use string functions
Impossible values	Manual overrules
Missing values	Remove observation or value
Outliers	Validate and, if erroneous, treat as missing value (remove or insert)
<i>Errors pointing to inconsistencies between data sets</i>	
Deviations from a code book	Match on keys or else use manual overrules
Different units of measurement	Recalculate
Different levels of aggregation	Bring to same level of measurement by aggregation or extrapolation

DDATA PREPARATION CLEANSING DATA

❑ Advanced methods to find and identify data errors: (rarely applied)

- Simple modeling
- Diagnostic plots
- Regression



DATA PREPARATION CLEANSING DATA

DATA ENTRY ERRORS

- ❑ Data collection and data entry are error-prone processes
- ❑ Human errors → typos, because of lack of concentration or sloppiness
- ❑ Machine errors → machine or h/w failure or transmission errors – eg. during ETL
- ❑ Detect errors by tabulating the frequency counts (for different classes) – then use simple assignment statements and if-then-else rules to fix them

Value	Count
Good	1598647
Bad	1354468
Godo	15
Bade	1

Frequency Table

```
if x == "Godo":
    x = "Good"
if x == "Bade":
    x = "Bad"
```

DATA PREPARATION CLEANSING DATA

REDUNDANT WHITESPACE

- ❑ Hard to detect but cause errors
- ❑ Takes time to identify the errors – hence causes delay in the process
- ❑ Eg. 'MR' is different from 'MR '.
- ❑ Fix using strip() function in Python (to remove leading and trailing spaces)

FIXING CAPITAL LETTER MISMATCHES

- ❑ Convert all letters to lowercase and use the data (using .lower() in Python)

DATA PREPARATION CLEANSING DATA

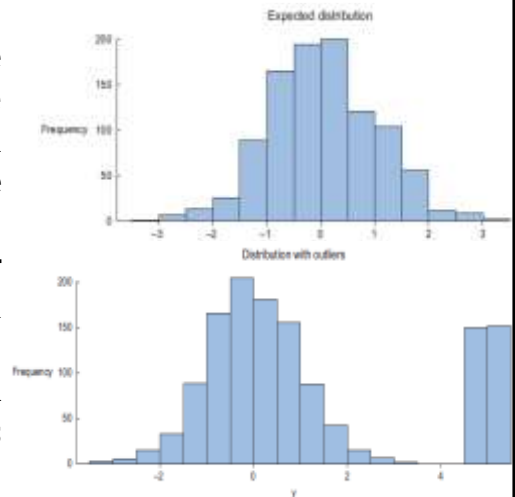
IMPOSSIBLE VALUES AND SANITY CHECKS

- ❑ Check the value against physically or theoretically impossible values – eg. age of a person > 200
- ❑ Check using rules: eg. check = $0 \leq \text{age} \leq 120$

DATA PREPARATION CLEANSING DATA

OUTLIERS

- ❑ An outlier → an observation that seems to be distant from other observations or, more specifically, one observation that follows a different logic or generative process than the other observations.
- ❑ The easiest way to find outliers → use a plot or a table with the minimum and maximum values.
- ❑ Outliers can seriously influence the data modeling, so investigating them first is essential



DATA PREPARATION CLEANSING DATA

DEALING WITH MISSING VALUES

- ❑ Must be handled necessarily and separately
- ❑ May indicate that something went wrong in data collection or during ETL process.
- ❑ **Techniques to handle missing data:**

Technique	Advantage	Disadvantage
Omit the values	Easy to perform	You lose the information from an observation
Set value to null	Easy to perform	Not every modeling technique and/or implementation can handle null values
Impute a static value such as 0 or the mean	Easy to perform You don't lose information from the other variables in the observation	Can lead to false estimations from a model
Impute a value from an estimated or theoretical distribution	Does not disturb the model as much	Harder to execute You make data assumptions
Modeling the value (nondependent)	Does not disturb the model too much	Can lead to too much confidence in the model Can artificially raise dependence among the variables Harder to execute You make data assumptions

DATA PREPARATION CLEANSING DATA

DEVIATIONS FROM A CODE BOOK

- ❑ A code book is a description of the data, a form of metadata (data about data)
- ❑ It contains
 - The number of variables per observation
 - The number of observations
 - What each encoding within a variable means (eg. “0” equals “negative”, “5” stands for “very positive”).
 - The type of data you’re looking at (hierarchical data or graph or image, etc.)
 - Use SETS to check whether the data is as per the code book.

DATA PREPARATION CLEANSING DATA

DIFFERENT UNITS OF MEASUREMENT

- ❑ When integrating two data sets, pay attention to their respective units of measurement.
- ❑ Eg. Studying the prices of oil in the world → gather data from different data providers → data sets can contain prices per gallon and others can contain prices per liter.
- ❑ A simple conversion will be required to fix the differences.

DIFFERENT LEVELS OF AGGREGATION

- Eg. A data set containing data per week and another dataset containing data per work week.
- This type of error is easy to detect
- Summarizing (or the inverse, expanding) the data sets will fix it.

DATA PREPARATION

CORRECT ERRORS AS EARLY AS POSSIBLE

- ❑ Fixing the data as soon as it's captured is nice in a perfect world.
- ❑ Data should be cleansed when acquired for the following reasons:
 - **Not everyone spots** the data anomalies. Decision-makers may make costly mistakes on information based on incorrect data from applications that fail to correct for the faulty data.
 - If errors are not corrected early on in the process, the cleansing will have to be done **for every project** that uses that data.
 - Data errors may **point to a business process that isn't working as designed**. Eg. clients abusing the couponing system and earned money while purchasing groceries
 - Data errors may **point to defective equipment**, such as broken transmission lines and defective sensors.
 - Data errors can **point to bugs in software** or in the integration of software that may be critical to the company. Eg. different applications with different local settings used by the same organization at different locations

DATA PREPARATION

COMBINING DATA FROM DIFFERENT DATA SOURCES

- ❑ Data comes from several different places
- ❑ Integrating these different sources has to be done.
- ❑ Data varies in size, type, and structure, ranging from databases and Excel files to text documents.

THE DIFFERENT WAYS OF COMBINING DATA

- *Joining*: enriching an observation from one table with information from another table.
- *Appending* or *stacking*: adding the observations of one table to those of another table.
- ❑ Create a new physical table or a virtual table by creating a view (doesn't consume more disk space).

DATA PREPARATION

COMBINING DATA FROM DIFFERENT DATA SOURCES

JOINING TABLES

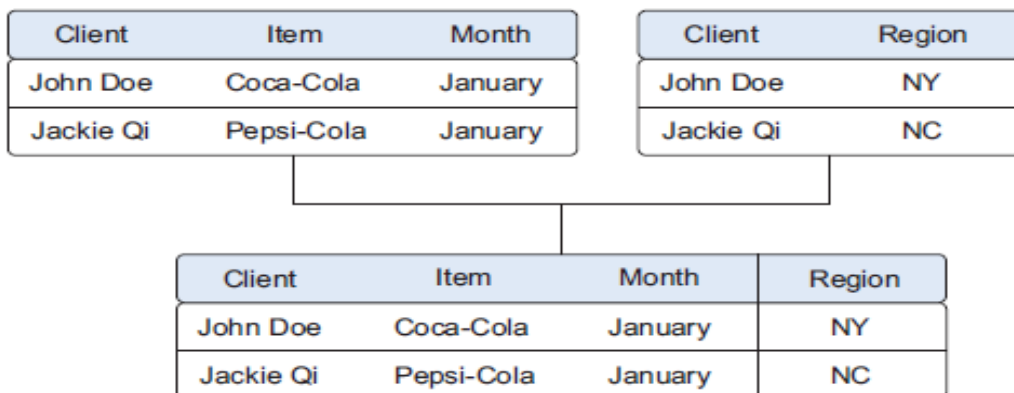
- ❑ To combine the information of one observation found in one table with the information that you find in another table → focusing on enriching a single observation.
- Use variables that represent the same object in both tables, such as a date, a country name, or a Social Security number → these common fields are known as keys.
- Primary keys → When these keys also uniquely define the records in the table (Eg. Aadhaar no., PAN)
- Join → Using *lookup* function in Excel

DATA PREPARATION

COMBINING DATA FROM DIFFERENT DATA SOURCES

JOINING TABLES

- ❑ Example



DATA PREPARATION

COMBINING DATA FROM DIFFERENT DATA SOURCES

APPENDING TABLES

- ❑ Adding observations from one table to another table.
- ❑ The result of appending two tables is a larger one with the observations from the 1st as well as the 2nd one.
- ❑ Eg. Consider 2 tables with data from January and February months → they can be combined.
- ❑ Similar to union operation
- ❑ But requires an equal structure in the tables being appended

DATA PREPARATION

COMBINING DATA FROM DIFFERENT DATA SOURCES

APPENDING TABLES

- ❑ Example

Client	Item	Month
John Doe	Coca-Cola	January
Jackie Qi	Pepsi-Cola	January

Client	Item	Month
John Doe	Zero-Cola	February
Jackie Qi	Maxi-Cola	February

Client	Item	Month
John Doe	Coca-Cola	January
Jackie Qi	Pepsi-Cola	January
John Doe	Zero-Cola	February
Jackie Qi	Maxi-Cola	February

DATA PREPARATION

COMBINING DATA FROM DIFFERENT DATA SOURCES

USING VIEWS TO SIMULATE DATA JOINS AND APPENDS

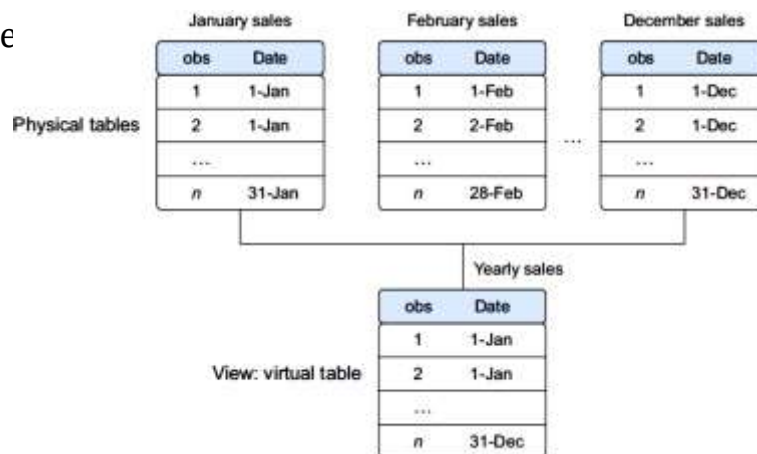
- ❑ Joining or appending may cause duplication of data when combined in a new physical table – occupies storage space.
- ❑ This issue can be overcome by 'View'.
- ❑ View → like a table → a virtual layer that combines the tables
- ❑ Advantage: saves space
- ❑ Disadvantage: uses more processing power → because the join that creates the view is recreated every time it's queried

DATA PREPARATION

COMBINING DATA FROM DIFFERENT DATA SOURCES

VIEWS

- ❑ Example



DATA PREPARATION

COMBINING DATA FROM DIFFERENT DATA SOURCES

ENRICHING AGGREGATED MEASURES

- ❑ Data enrichment can also be done by adding calculated information to the table
- ❑ Eg. total number of sales or percentage of total stock sold in a certain region.
- ❑ Could be useful during data exploration but more so when creating data models.

Product class	Product	Sales in \$	Sales t-1 in \$	Growth	Sales by product class	Rank sales
A	B	X	Y	$(X-Y) / Y$	AX	NX
Sport	Sport 1	95	98	-3.06%	215	2
Sport	Sport 2	120	132	-9.09%	215	1
Shoes	Shoes 1	10	6	66.67%	10	3

DATA PREPARATION

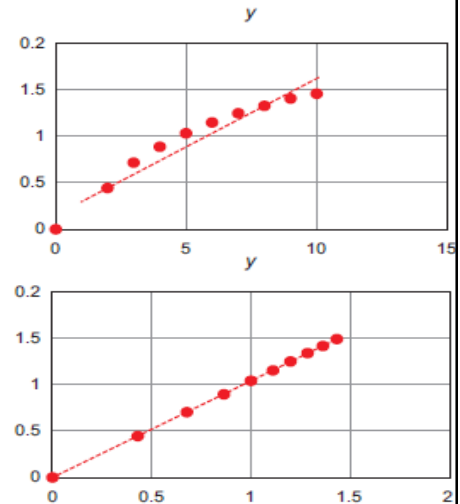
TRANSFORMING DATA

- ❑ Certain models require their data to be in a certain shape (no. of rows, no. of columns, etc.)
- ❑ So any available raw data has to be converted into a form suitable for data modelling.
- ❑ Transforming data → the process of converting raw data into a format that is suitable for analysis and modeling.
- ❑ Purpose → to prepare the data for data science so that the data can be used to extract useful insights and knowledge.

TRANSFORMING DATA

TRANSFORMING DATA FOR LINEARITY

- ❑ To make the relationships between an input variable and an output variable linear.
- ❑ By combining two variables into a new variable.
- ❑ Eg. Let $y = ae^{bx}$
- ❑ When $x = 1, 2, 3, \dots, 10$, the relationship between x and y is non-linear.
- ❑ When $x = 1, 2, 3, \dots, 10$, and when x is transformed into $\log(x)$, the relationship between $\log(x)$ and y is transformed into a linear one.



TRANSFORMING DATA

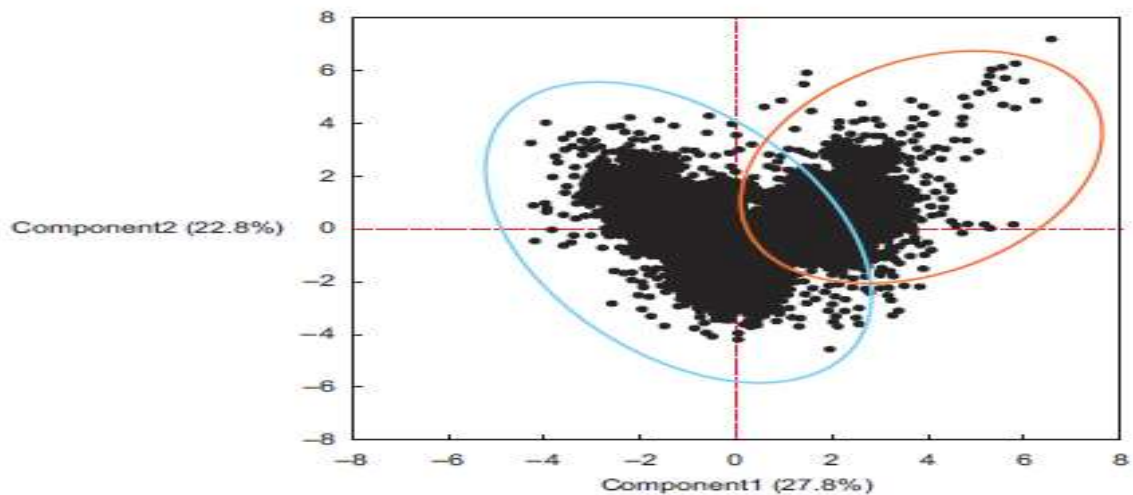
REDUCING THE NUMBER OF VARIABLES

- ❑ Too many variables - Need to be reduced using special methods because
 - They don't add new information to the model
 - Makes the model difficult to handle
 - Certain techniques don't perform well when overloaded with too many input variables
 - Helps to understand the important values
- ❑ Eg. Methods using Euclidean distance perform well only up to 10 variables.
- ❑ Reducing variables must retain the maximum amount of data
- ❑ Let Component 1 = combination of variables 1 to 5
- ❑ Let Component 2 = combination of variables 6 to 8
- ❑ Principal components $\rightarrow$ variables in component 1
- ❑ Analysis on them $\rightarrow$ Principal Components Analysis (PCA)

$$= \sqrt{((x1-x2)^2 + (y1-y2)^2)}$$

TRANSFORMING DATA

REDUCING THE NUMBER OF VARIABLES



TRANSFORMING DATA

TURNING VARIABLES INTO DUMMIES

- ☐ Dummy variables – can take only take two values: true(1) or false(0)
- ☐ Used to indicate the absence of a categorical effect that may explain the observation.
- ☐ Make separate columns for the classes stored in one variable.
- ☐ Indicate it with 1 if the class is present and 0 otherwise.
- ☐ Eg. turning one column named Weekdays into the columns Monday through Sunday.
- ☐ Eg. turning one column named Gender into the columns Male and Female.

TRANSFORMING DATA TURNING VARIABLES INTO DUMMIES

Eg. 1

Customer	Year	Gender	Sales
1	2015	F	10
2	2015	M	8
1	2016	F	11
3	2016	M	12
4	2017	F	14
3	2017	M	13

M F

Customer	Year	Sales	Male	Female
1	2015	10	0	1
1	2016	11	0	1
2	2015	8	1	0
3	2016	12	1	0
3	2017	13	1	0
4	2017	14	0	1

Eg. 2

ID	WEEKDAY
103	MONDAY
362	THURSDAY
730	TUESDAY

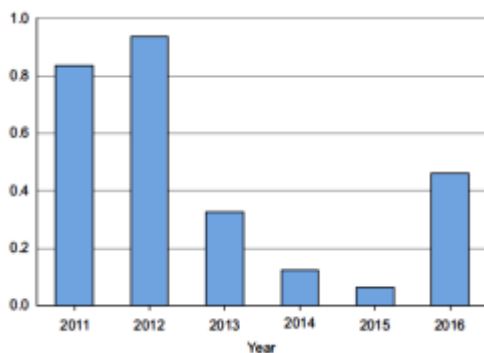
ID	M	T	W	TH	F	SAT	SUN
103	1	0	0	0	0	0	0
362	0	0	0	1	0	0	0
730	0	1	0	0	0	0	0

4. EXPLORATORY DATA ANALYSIS

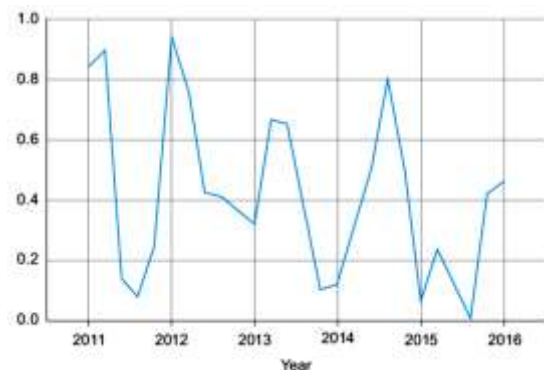
EXPLORATORY DATA ANALYSIS

- ❑ Taking a deep dive into the data to explore data
- ❑ Use graphical techniques to gain an understanding of your data and the interactions between variables
- ❑ Data cleaning completed - Still anomalies missed before may be discovered - and fix them and proceed
- ❑ Visualization techniques - simple line graphs or histograms
- ❑ Use composite graph from simple graphs
- ❑ Use animated and interactive graphs – eg. Sankey diagram
- ❑ Not limited to visualization techniques.
- ❑ Other techniques - Tabulation, clustering, and other simple modeling techniques.

EXPLORATORY DATA ANALYSIS

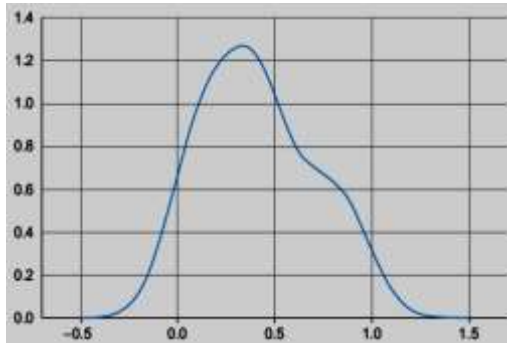


Bar chart

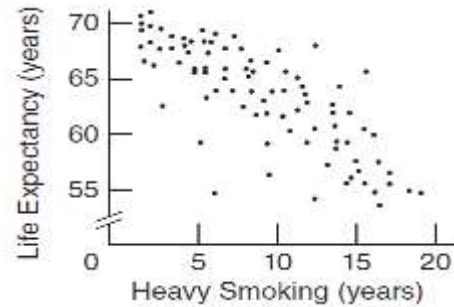


Line plot

EXPLORATORY DATA ANALYSIS



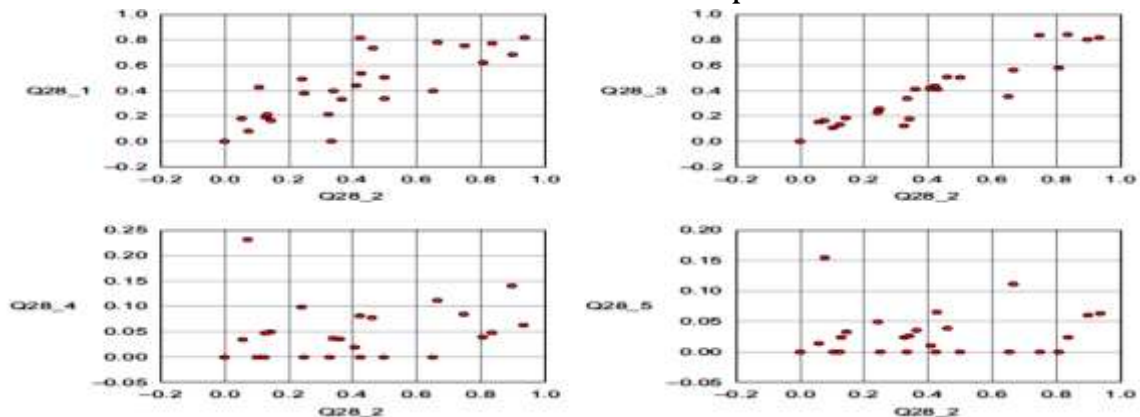
Distribution



Scatterplot

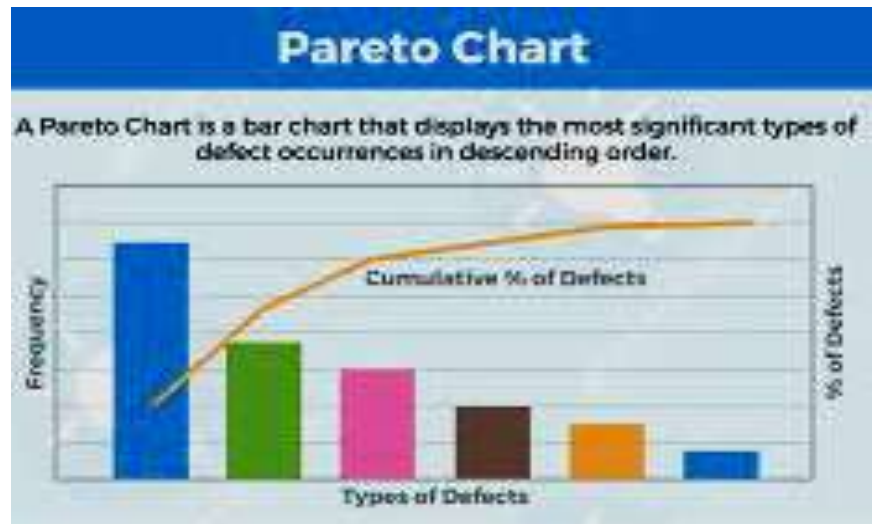
EXPLORATORY DATA ANALYSIS

Combining multiple plots - to provide even more insight – helps to understand the structure of the data over multiple variables



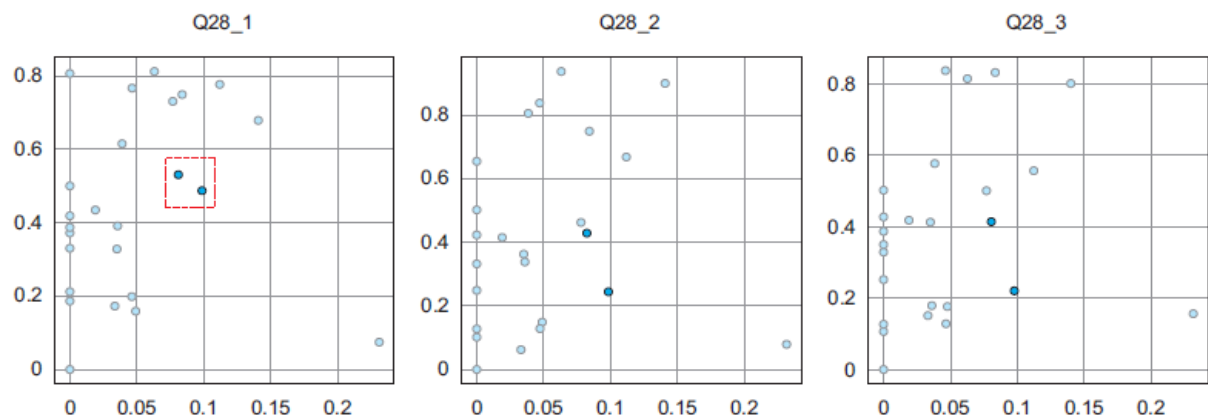
EXPLORATORY DATA ANALYSIS

- Overlaying several plots
- simple graphs combined into a Pareto diagram, or 80-20 diagram
- a combination of the values and a cumulative distribution.



EXPLORATORY DATA ANALYSIS

Brushing and linking - select observations in one plot and highlight the same observations in the other plots

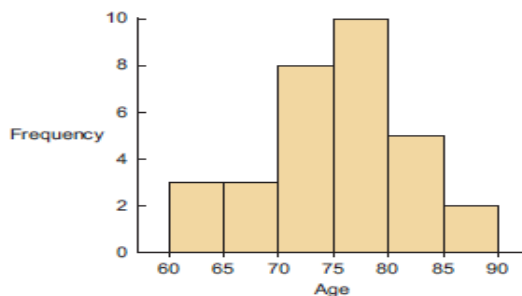


EXPLORATORY DATA ANALYSIS

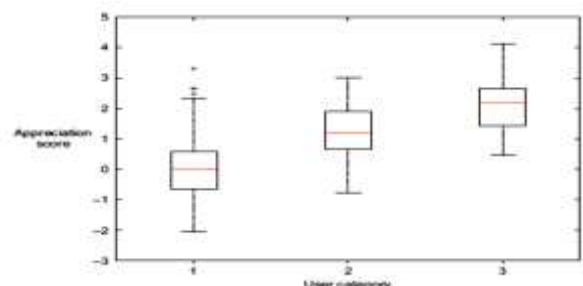
- ❑ Brushing and linking - select observations in one plot and highlight the same observations in the other plots
- ❑ Combines and links different graphs and tables (or views) so changes in one graph are automatically transferred to the other graphs.
- ❑ An interactive exploration of data that facilitates the discovery of new insights.

EXPLORATORY DATA ANALYSIS

Histogram - a variable is cut into discrete categories and the number of occurrences in each category are summed up and shown in the graph.



Boxplot - offers an impression of the distribution within categories - shows the maximum, minimum, median, and other characterizing measures at the same time.



5. BUILD THE MODELS

MODEL AND VARIABLE SELECTION

MODEL EXECUTION

MODEL DIAGNOSTICS AND MODEL COMPARISON

BUILD THE MODELS

- ❑ A machine learning model is defined as a mathematical representation of the output of the training process - a program that has been trained to find patterns within new data and make predictions.
- ❑ The goal of building models - making better predictions, classifying objects, or gaining an understanding of the system that you're modeling.
- ❑ More focused phase in data science process – because data scientists know what outcome is wanted.
- ❑ Iterative process
- ❑ Components:
 - Model and variable selection
 - Model execution
 - Model diagnostic and model comparison

BUILD THE MODELS

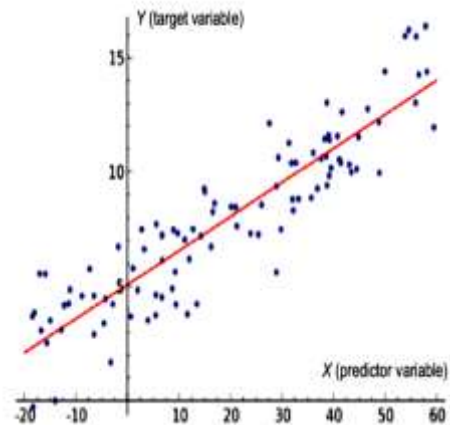
- ❑ Techniques used are borrowed from the field of machine learning, data mining, and statistics.
- ❑ Main steps in the techniques:
 1. Selection of a modeling technique and variables to enter in the model
 2. Execution of the model
 3. Diagnosis and model comparison
- ❑ Select the variables you want to include in your model and a modeling technique
- ❑ Findings from the exploratory analysis (set of variables to construct a good model) will help.
- ❑ Choosing the right model for a problem is essential.

1. MODEL AND VARIABLE SELECTION

- ❑ Consider
 - model performance
 - whether your project meets all the requirements to use the model
 - Other factors:
 - ✓ Must the model be moved to a production environment and, if so, would it be easy to implement?
 - ✓ How difficult is the maintenance on the model: how long will it remain relevant if left untouched?
 - ✓ Does the model need to be easy to explain?

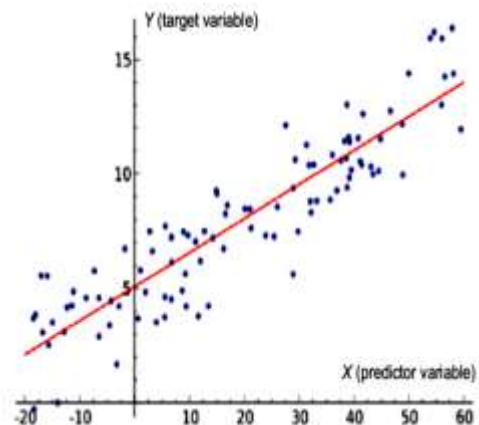
2. MODEL EXECUTION

- ❑ Implement the model in code.
- ❑ Use
 - Python libraries such as StatsModels or Scikit-learn - can speed up the process.
 - R – a free software environment, widely used for statistical computing.
- ❑ For eg. to predict a value → use **linear regression**
- ❑ In linear regression, y (target variable) is computed based on x (predictor variable) to get the linear relationship / between x and y and the dependence of y on x .



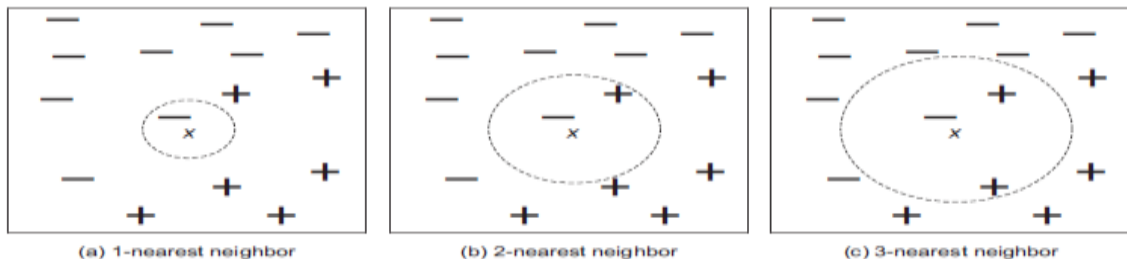
2. MODEL EXECUTION

- ❑ The remarks on output:
 - a) Model fit: R-squared or adjusted R-squared is used to fit the regression model.
 - b) Predictor variables have a coefficient
 - c) Predictor significance



2. MODEL EXECUTION

- ❑ To classify a set of values – use k-nearest neighbors (KNN)
- ❑ k-nearest neighbors looks at labeled points nearby an unlabeled point and, based on this, makes a prediction of what the label should be.



3. MODEL DIAGNOSTICS AND MODEL COMPARISON

- ❑ Build multiple models - from them choose the best one based on multiple criteria.
- ❑ Working with a holdout sample - helps you pick the best-performing model.
- ❑ A holdout sample → a part of the data you leave out of the model building so it can be used to evaluate the model afterward.
- ❑ Split 80% of data as “training dataset” and remaining 20% as “test dataset” - Use model on the unseen data (test data) and calculate error measures to evaluate it.
- ❑ Error measure – used to check for every prediction how far it differs from the truth

3. MODEL DIAGNOSTICS AND MODEL COMPARISON

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\hat{Y}_i - Y_i)^2$$

- ❑ An example: Mean Squared Error
- ❑ For multiple models - choose the model with the lowest error

	<i>n</i>	Size	Price	Predicted model 1	Predicted model 2	Error model 1	Error model 2
80% train	1	10	3				
	2	15	5				
	3	18	6				
	4	14	5				
	...	...					
	800	9	3				
	801	12	4	12	10	0	2
	802	13	4	12	10	1	3
	...						
	999	21	7	21	10	0	11
20% test	1000	10	4	12	10	-2	0
Total						5861	110225

3. MODEL DIAGNOSTICS AND MODEL COMPARISON

- Model diagnostics → Many models make strong assumptions (such as independence of the inputs) and you have to verify that these assumptions are met.

6. PRESENTING FINDINGS AND BUILDING APPLICATIONS ON TOP OF THEM



PRESENTING FINDINGS AND BUILDING APPLICATIONS ON TOP OF THEM

- ❑ Explain what you found to the stakeholders.
- ❑ Sometimes repetition needed – automate your models - to avoid redoing all of your analysis all the time.
- ❑ Build an application that automatically updates reports, Excel spreadsheets, or PowerPoint presentations.
- ❑ Apply soft skills (!)

DATA SCIENCE PROCESS – A SUMMARY

- ❑ **Setting the research goal** → Defining the what, the why, and the how of your project in a project charter.
- ❑ **Retrieving data** → Finding and getting access to data needed in your project. This data is either found within the company or retrieved from a third party.
- ❑ **Data preparation** → Checking and remediating data errors, enriching the data with data from other data sources, and transforming it into a suitable format for your models.
- ❑ **Data exploration** → Diving deeper into your data using descriptive statistics and visual techniques.
- ❑ **Data modeling** → Using machine learning and statistical techniques to achieve your project goal.
- ❑ **Presentation and automation** → Presenting your results to the stakeholders and industrializing your analysis process for repetitive reuse and integration with other tools.

BASIC STATISTICAL DESCRIPTIONS OF DATA

STATISTICAL DESCRIPTIONS OF DATA

- ❑ For data preprocessing to be successful, it is essential to have an overall picture of our data.
- ❑ Basic statistical descriptions can be used to identify properties of the data and highlight which data values should be treated as noise or outliers.
- ❑ For data preprocessing tasks, we want to learn about data characteristics regarding both central tendency and dispersion of the data.
- ❑ Measures of central tendency (descriptive summary of a dataset) include **mean**, **median** and **mode**.
- ❑ Measures of data dispersion include **variance**, **standard deviation**, **quartiles** and **interquartile range (IQR)**.
- ❑ These descriptive statistics are of great help in understanding the distribution of the data.

STATISTICAL DESCRIPTIONS OF DATA

MEAN

- ❑ The mean of a data set $\rightarrow$ the average of all the data values.
- ❑ The sample mean $\bar{x}$ $\rightarrow$ the point estimator of the population mean μ .

$$\text{Sample mean } \bar{x} = \frac{\text{Sum of the values of then observations}}{\text{Number of observations in the sample}} = \frac{\sum x_i}{n}$$

$$\text{Population mean } \mu = \frac{\text{Sum of the values of the N observations}}{\text{Number of observations in the population}} = \frac{\sum x_i}{n}$$

STATISTICAL DESCRIPTIONS OF DATA

MEDIAN

- ❑ The median of a data set → the value in the middle when the data items are arranged in ascending order.
- ❑ Whenever a data set has extreme values, the median is the preferred measure of central location.
- ❑ For an odd number of observations:
 - 7 observations = 26, 18, 27, 12, 14, 29, 19.
 - Arrange numbers in ascending order = 12, 14, 18, 19, 26, 27, 29
 - The median is the middle value: Median = 19
- ❑ For an even number of observations:
 - 8 observations = 26, 18, 29, 12, 14, 27, 30, 19
 - Arrange numbers in ascending order = 12, 14, 18, **19, 26**, 27, 29, 30
 - The median is the average of the middle two values: Median = $45/2 = 22.5$

STATISTICAL DESCRIPTIONS OF DATA

MODE

- ❑ The mode of a data set → the value that occurs with greatest frequency.
- ❑ The greatest frequency can occur at two or more different values.
- ❑ If the data have exactly two modes, the data are bimodal.
- ❑ If the data have more than two modes, the data are multimodal.

STATISTICAL DESCRIPTIONS OF DATA QUARTILES

- ❑ Quartiles → three values that split sorted data into four parts, each with an equal number of observations.
- ❑ First quartile (Q1): The first quartile is the value, where 25% of the values are smaller than Q1 and 75% are larger.
- ❑ Second quartile (Q2): The second quartile is the value, where 50% of the values are smaller than Q2 and 50% are larger. Q2 is the median.
- ❑ Third quartile (Q3): The third quartile is the value, where 75 % of the values are smaller than Q3 and 25% are larger.

STATISTICAL DESCRIPTIONS OF DATA VARIANCE

- ❑ The variance is the average of the squared difference between each data value and the mean.
- ❑ It is a measurement of the spread between numbers in a data set.
- ❑ It is based on the difference between the value of each observation (x_i) and the mean ($\bar{x}$ for a sample, μ for a population).

$$\text{Sample variance : } S^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

$$\text{Population variance : } \sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

STATISTICAL DESCRIPTIONS OF DATA

STANDARD DEVIATION

- ❑ The standard deviation of a data set → the positive square root of the variance.
- ❑ It is a measure which shows how much variation from the mean exists.
- ❑ It is measured in the same in the same units as the data, making it more easily interpreted than the variance.

Population standard deviation = σ

$$= \sqrt{\sigma^2}$$

$$= \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

Sample standard deviation = S

$$= \sqrt{S^2}$$

$$= \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

STATISTICAL DESCRIPTIONS OF DATA

End of Unit 1

- ❑ **Example:**
- ❑ Find the mean, median, mode, variance and standard deviation for the following data
2, 17, 5, 3, 28, 7, 5, 8, 5, 6, 2, 12, 10, 4, 3.

